6LoWPAN: IPv6 in Networks of Constrained Devices

Communication Technologies for the Internet of Things, Lecture 09

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Agenda (1/2)

- Networks of constrained devices: the motivation for IP and IPv6
- The problem of header size
- The 6LoWPAN adaptation layer
- Header compression
- Fragmentation and reassembly
- Stateless address autoconfiguration

Agenda (2/2)

- From 6LoWPAN to 6Lo: extension to other technologies
- A real project example and its constraints

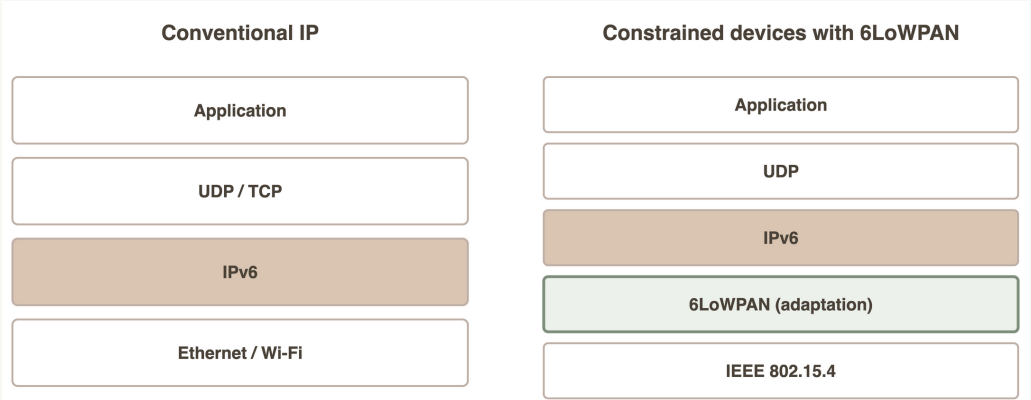
What 6LoWPAN Is (1/2)

- It designates the transport of IPv6 in low-power wireless personal area networks, built on the IEEE 802.15.4 standard.
- It is an adapted version of IPv6, with reduced headers and a simplified form of the protocol.
- It corresponds to a series of IETF specifications, started in 2007.
- It works as an adaptation layer between IPv6 and IEEE 802.15.4 links, resolving the incompatibilities between the two.

What 6LoWPAN Is (2/2)

- It allows direct IP addressing of nodes, with no need for translation by a proprietary gateway.
- It uses the unslotted CSMA/CA mode of IEEE 802.15.4.

6LoWPAN in the Protocol Stack



The adaptation layer compresses, fragments and bridges the gap between IPv6 and 802.15.4

Why Carry IP over 802.15.4 (1/2)

- The ubiquity of IP networks makes it possible to reuse the installed infrastructure instead of building a parallel one.
- IP-based technologies have existed for decades, are well understood and are proven to work at scale.
- The specifications are open and freely available, unlike the closed proprietary solutions that dominated sensor networks.

Why Carry IP over 802.15.4 (2/2)

- Mature diagnostic, management and configuration tools already exist for IP networks, and can now be applied to sensor networks.
- The alternative (proprietary protocols with translation at a gateway) means maintaining, for each technology, its own ecosystem of tools and skills.

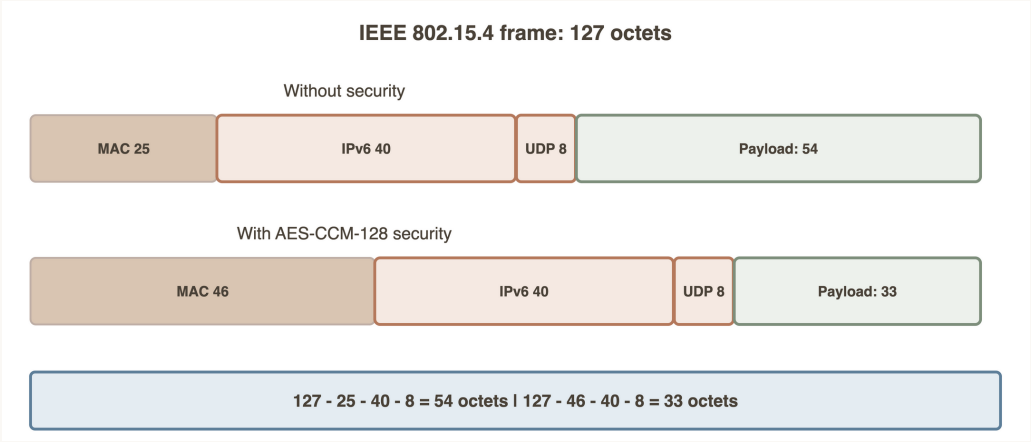
The Header Size Problem (1/2)

- The IPv6 header takes 40 octets and the UDP header takes 8 octets, a total of 48 octets.
- The IEEE 802.15.4 MAC header can take up to 25 octets without security, or 46 octets with AES-CCM-128 security.
- The IEEE 802.15.4 frame has a maximum size of 127 octets.
- Doing the sums without security: $127 - 25 - 40 - 8$ leaves only 54 octets for the application payload.

The Header Size Problem (2/2)

- With AES-CCM-128 security: $127 - 46 - 40 - 8$ leaves only 33 octets.
- The ratio of header information to application payload is, under these conditions, plainly poor, and this is the problem 6LoWPAN exists to solve.

The Budget of a 127-Octet Frame



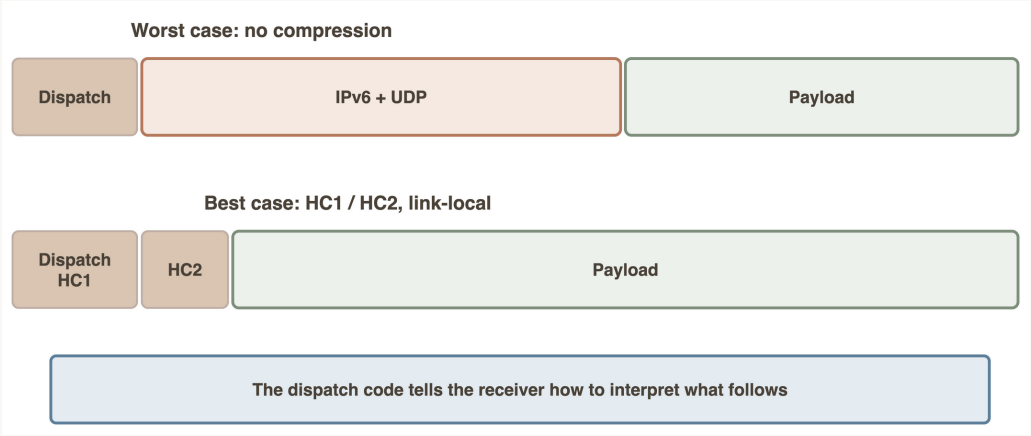
Dispatch Codes and Frame Formats (1/2)

- All datagrams encapsulated in 6LoWPAN are preceded by a stack of encapsulation headers.
- Each header in the stack begins with a type field, followed by zero or more fields of its own.
- This type field, called the dispatch code, tells the receiver how to interpret what follows.
- **Worst case, uncompressed IPv6/UDP:** the dispatch code indicates no compression, leaving 54 or 33 octets for the payload, depending on the security used.

Dispatch Codes and Frame Formats (2/2)

- **Best case, compressed link-local IPv6/UDP:** the dispatch code indicates HC1 compression, which may in turn indicate that HC2 compression follows.
- This second case is the maximum compression achievable, but only for local-scope addresses; it does not work for global addresses.
- Header fields that cannot be compressed are carried after the HC1 or HC1/HC2 marks, in a partial compression scheme.

Worst and Best Encapsulation Case



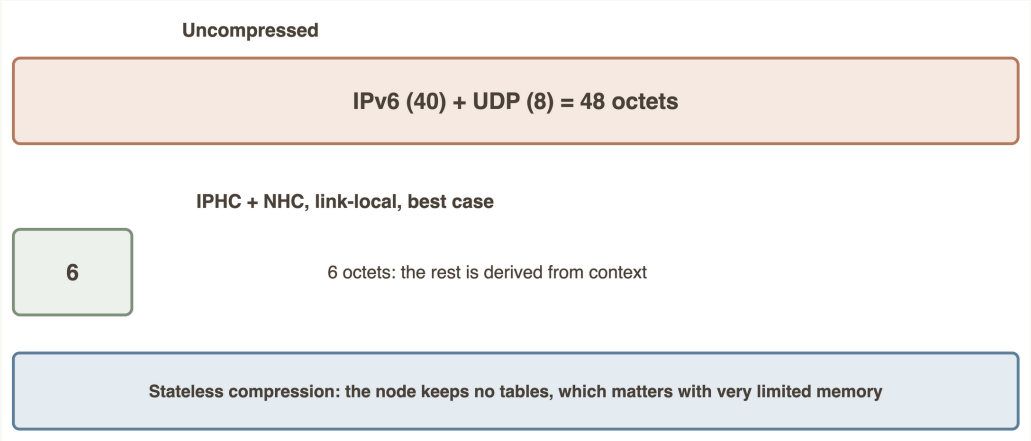
Header Compression Principles (1/2)

- The compression principles are defined in RFC 4944.
- **First principle:** omit all header fields that can be derived from context, and send the rest unchanged.
- **Second principle:** nodes do not have to keep compression state, which makes the compression stateless. This choice is decisive on devices with very limited memory.
- **Third principle:** the scheme supports practically arbitrary combinations of compressed and uncompressed fields.

Header Compression Principles (2/2)

- Later work extended the scheme to the compression of globally routable addresses and introduced stateful compression, through the IPHC and NHC mechanisms.
- With IPHC and NHC applied to link-local IPv6 with UDP, the 48 bytes of header can, in the best case, be reduced to just 6 bytes.

From 48 Octets to 6



Generic Header Compression (RFC 7400) (1/2)

- The original scheme required a new compression mechanism to be defined for each new type of header to be compressed, which was not sustainable as protocols multiplied.
- Generic header compression, defined in RFC 7400, solves this problem by adding a much more general, if slightly less efficient, scheme.
- It is integrated into the existing NHC mechanism.

Generic Header Compression (RFC 7400) (2/2)

- It is based on LZ-77 style compression, with opcodes that allow zeros to be appended, entries of a static dictionary to be referenced, or data to be copied unchanged.
- Support for generic compression is announced through Neighbor Discovery options.

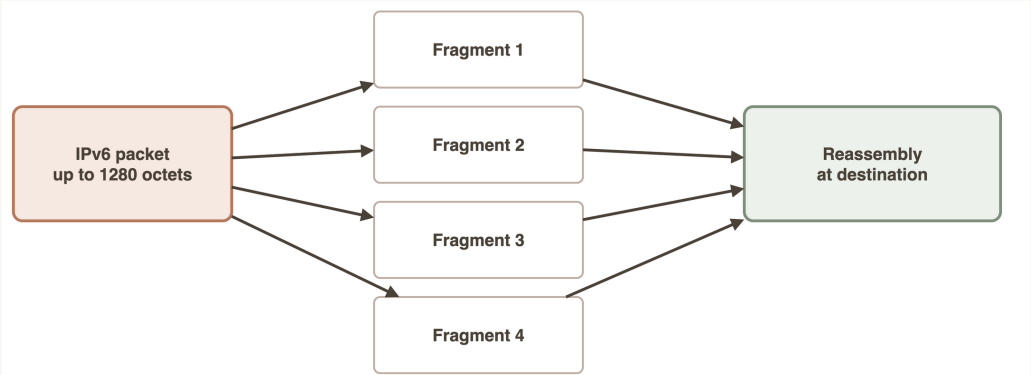
Fragmentation and Reassembly (1/2)

- IPv6 accepts packets of up to 1280 bytes.
- This size cannot be accommodated in a maximum transmission unit of 127 bytes, such as that of IEEE 802.15.4.
- 6LoWPAN therefore defines a fragmentation scheme that allows such packets to be carried.
- The fragmentation header, with 11 bits, supports packets up to 2048 bytes in total size.
- The cost is a multiplication of the number of fragments in transit.

Fragmentation and Reassembly (2/2)

- Fragmentation degrades performance in lossy networks: losing a single fragment forces the whole datagram to be retransmitted, so the probability of success falls quickly with the number of fragments.
- Hence the practical recommendation: avoid large packets whenever the application design allows.

Fragmenting an IPv6 Packet



Losing one fragment forces retransmission of the whole datagram

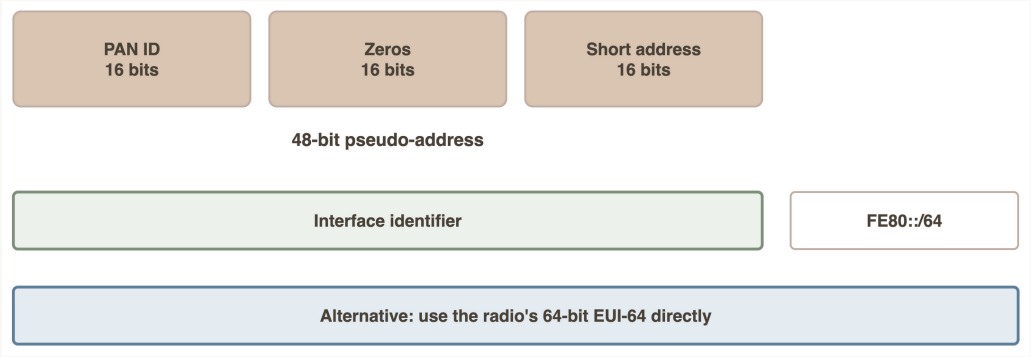
Stateless Address Autoconfiguration (1/2)

- Autoconfiguration is based on the device's layer-2 address, as in conventional IPv6.
- A hardware EUI-64 address can be used directly as the interface identifier.
- Alternatively, a 48-bit pseudo-address can be built from the short address, combining the 16-bit PAN identifier, 16 zero bits and the 16-bit short address.
- Local-scope addresses use the prefix FE80::/64, as in any IPv6 network.

Stateless Address Autoconfiguration (2/2)

- Neighbor Discovery adapted to 6LoWPAN is specified in RFC 6775, which introduces the optimisations needed in low-power networks.
- These optimisations are essential because conventional Neighbor Discovery relies on frequent multicast, which is incompatible with nodes that spend most of their time asleep.

From the Short Address to the Interface Identifier



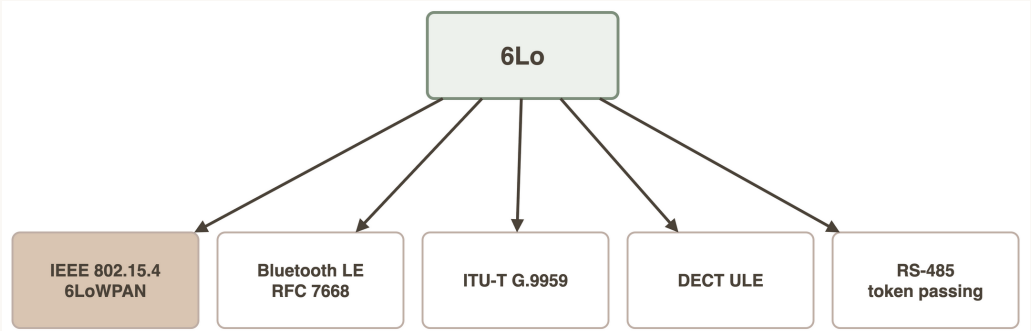
From 6LoWPAN to 6Lo (1/2)

- The 6LoWPAN adaptation layer was declared complete at the end of 2012, and its working group was closed.
- Meanwhile, the variety of constrained devices kept growing, and with it the need to support IPv6 over other layer-2 technologies.
- In 2013, the IETF created the IPv6 over Networks of Resource-constrained Nodes working group, known as 6Lo, building on the work already done in 6LoWPAN.
- The scope of 6Lo covers Bluetooth Low Energy, the ITU-T G.9959 standard, DECT Ultra Low Energy and master-slave token-passing networks over RS-485.

From 6LoWPAN to 6Lo (2/2)

- The transport of IPv6 over Bluetooth Low Energy is specified in RFC 7668: it dispenses with fragmentation, but keeps the compression methods.
- On Linux, this translates into shared common code between the wireless personal area network and Bluetooth subsystems.
- Other adaptations in preparation include NFC, power-line communications and IEEE 802.11ah.

The Scope of 6Lo



Same compression methods over other link layers

Reference Implementations and Specifications (1/2)

- On Linux, support is provided by the Linux-WSN project, which offers raw sockets for IEEE 802.15.4 and 6LoWPAN drivers.
- Supported hardware includes the Atmel AT86RF230 and Microchip MRF24J40 transceivers.
- Known limitations include incomplete support for all address compression types, which may compromise interoperability with other vendors' implementations.

Reference Implementations and Specifications (2/2)

- **Reference specifications:** RFC 4919 states the problem; RFC 4944 defines the transmission of IPv6 packets over IEEE 802.15.4 networks; RFC 6282 defines the compression format for IPv6 datagrams; RFC 6550 defines the RPL routing protocol; RFC 6775 defines the Neighbor Discovery optimisations; RFC 7400 defines generic header compression.
- RPL, mentioned here, is the subject of the next lecture.

Case Study: Requirements of a Real Network (1/2)

- The scenario is the monitoring of a herd, with mobile sensors mounted on collars.
- The animals gather in small spaces, being gregarious, with herds that can reach 1500 head.
- The devices are very constrained, with little memory and low processing capacity.
- Communications are periodic, with a monitoring cycle of 60 seconds.

Case Study: Requirements of a Real Network (2/2)

- Energy saving is critical: more charge means more weight for the animal, and recharging means extra work for the shepherd.
- There are also fixed nodes, called beacons, used to establish a virtual fence based on received signal strength, and to collect the collars' measurements.

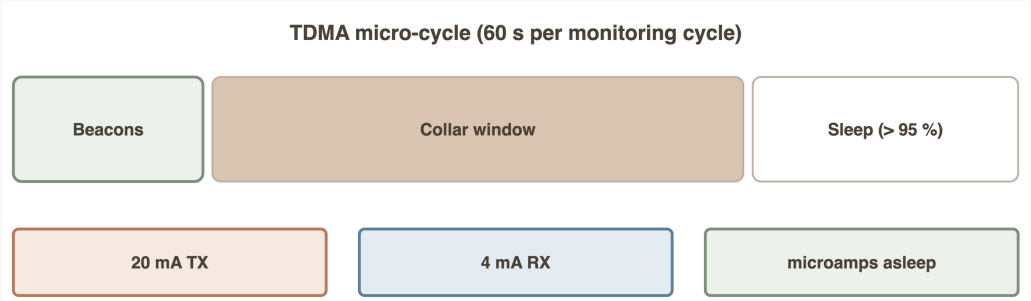
Case Study: Design Options (1/2)

- The chosen transceiver was the HopeRF RFM22, operating at 868 MHz, costing less than 3 euros, with a 64-byte MTU and a radio range of about 250 m.
- CSMA/CA medium access was rejected: the density of nodes would cause too many collisions, and each collision costs active radio time and therefore battery.
- TDMA communications were chosen, with radios going to sleep whenever they leave transmit and receive modes, spending more than 95% of the time asleep.
- Each micro-cycle starts with the beacons' communication, which lets the collars synchronise their clocks and measure signal strength for virtual-fence purposes.

Case Study: Design Options (2/2)

- Next comes the collars' window: each waits for its slot to communicate and goes back to sleep.
- The consumption figures illustrate the reason for these choices: 20 mA transmitting, 4 mA receiving and a few microamperes asleep.

The Case Study TDMA Cycle



No IP: 64-octet MTU, high density and the CPU busy analysing animal behaviour

Case Study: Critical Assessment (1/2)

- It was not possible to use IP or IPv6 in this project, and the reasons are instructive.
- The radio's MTU of only 64 bytes is half that of IEEE 802.15.4, which worsens the header problem already discussed.
- The node density in the infrastructure is very high, and CSMA/CA was ruled out because collisions cost awake radio time.
- 6TiSCH was also ruled out, because it requires processing capacity the device could not spare (the processor was needed to analyse animal behaviour) and because it adds cost.

Case Study: Critical Assessment (2/2)

- The price of these options is the loss of integration: without IP, the solution interoperates with nothing, and a full gateway is needed, not just a router.
- This is the lesson to retain: adopting IP on constrained devices is desirable, but it is neither free nor universally possible.

Lecture Summary (1/2)

- 6LoWPAN is an adaptation layer that makes it possible to carry IPv6 over IEEE 802.15.4 links, allowing direct IP addressing of constrained devices.
- The central problem is one of proportion: 48 bytes of IPv6 and UDP header in a 127-byte frame leave little room for useful data.
- Header compression, stateless by principle, solves much of the problem, and can reduce those 48 bytes to 6 in the best case.
- Fragmentation allows IPv6 packets of normal size to be carried, but degrades performance in lossy networks and should be avoided by design.

Lecture Summary (2/2)

- Stateless autoconfiguration and the optimised Neighbor Discovery defined in RFC 6775 let the network form without manual intervention.
- The work of 6LoWPAN was generalised by the 6Lo group to other technologies, including Bluetooth Low Energy.
- The next lecture covers routing in these networks, with 6TiSCH and the RPL, CoRPL and CARP protocols.

Thank you.

Questions and comments